

Eleven Primitives and Three Gates: The Universal Structure of Computational Imaging

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Abstract

Every computational imaging system—from a coded-aperture spectral camera to an MRI scanner—converts physical reality into measurements through a chain of physical transformations. Despite the apparent diversity of these systems across carrier families, length scales, and modalities, here we show that the space of all such forward models has *finite structural complexity*. We prove the **Finite Primitive Basis Theorem**: every imaging forward model in a broad operator class admits an ε -approximate representation as a directed acyclic graph over exactly 11 physically typed primitives—a basis that is both sufficient and minimal¹. This is a discovery, not a design choice: the primitive library saturates at 11 as modality coverage grows past 30 registered systems, with no new primitive required for the most recent 18 additions, proving that the structural complexity of imaging physics is bounded. We further establish the **Triad Decomposition**, a universal diagnostic law governing every failure in computational image recovery: reconstruction error decomposes into exactly three independent root causes—information deficiency, carrier noise, and operator mismatch—mirroring the role of entropy, thermal noise, and free-energy barriers in the thermodynamics of irreversible processes. Hardware validation across five real instruments spanning all four carrier families confirms that operator mismatch, not information deficiency or noise, is the dominant reconstruction bottleneck under standard operating conditions, recoverable by +0.8 to +10.7 dB (± 0.3 dB, 95% bootstrap CI) through forward-model correction alone—often exceeding the gain from upgrading a classical solver to a state-of-the-art deep network. These results shift the primary locus of investment in computational imaging from solver design to calibration, and provide a unifying blueprint for decomposing forward models across physical measurement science.

Introduction

Every imaging instrument—whether it captures spectra, spins, X-rays, or sound waves—converts a physical scene into digital measurements through a chain of transformations: a wave propagates, interacts with the object, passes through optics, and is recorded by a detector. Computational imaging reconstructs the scene by inverting this chain. The

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33 reconstruction algorithm assumes a mathematical model of the chain (the “forward model”),
 34 but this model inevitably diverges from reality: optics shift, detectors drift, and calibration
 35 degrades. When the model is wrong, the reconstruction fails—not because the algorithm is
 36 weak, but because it is solving the wrong problem.

37 This paper establishes that the space of imaging forward models has a surprisingly simple
 38 and universal structure, and that exploiting this structure reveals operator mismatch as
 39 the systematic, cross-modality bottleneck in reconstruction quality. Over the past decade,
 40 the community has invested enormous effort in improving reconstruction algorithms—
 41 progressing from compressed sensing^{2,3} and plug-and-play priors⁴ to deep unrolling net-
 42 works⁵ and vision transformers⁶—yet these algorithms routinely fail when deployed on
 43 real instruments⁷. Calibration methods exist for specific instruments—ESPIRiT⁸ for MRI,
 44 ePIE⁹ for ptychography—but they do not generalise across modalities, and no systematic
 45 framework decomposes reconstruction failures into root causes.

46 Here we establish two complementary foundations within Physics World Models (PWM).
 47 The **Finite Primitive Basis Theorem** (Theorem 3) proves that every imaging forward
 48 model in a broad operator class $\mathcal{C}_{\text{img}}$ admits an ε -approximate representation as a typed
 49 directed acyclic graph (DAG) over exactly 11 canonical primitives—and that this library is
 50 minimal¹. The **Triad Decomposition** proves that every reconstruction failure decomposes
 51 into three gates—information deficiency (**Gate 1**), carrier noise (**Gate 2**), and operator
 52 mismatch (**Gate 3**)—with a formal condition (Theorem 5) under which **Gate 3** dominates.
 53 The 11 primitives *enable* the 3-gate diagnosis: the DAG structure localizes mismatch to a
 54 specific physical operator, making correction actionable.

55 We validate both results across seven modalities spanning four carrier families—optical
 56 photons (CASSI¹⁰, CACTI¹¹, SPC¹², lensless), X-ray photons (CT¹³), electrons (ptychog-
 57 raphy¹⁴), and nuclear spins (MRI^{15,16})—with hardware validation on five real instruments
 58 confirming Triad predictions on physical measurements. A held-out closure test on 8 addi-
 59 tional modalities confirms basis completeness with saturating growth. The proof of Theo-
 60 rem 3 is in Supplementary Note 12; formal primitive semantics and typed DAG denotation
 61 are developed in¹.

62 **Why this matters beyond computational imaging.** Any physical measurement system—
 63 seismographs, telescopes, particle detectors, environmental sensor networks—faces the same
 64 structural problem: a forward model maps the quantity of interest to observations, and
 65 model fidelity limits what can be recovered. The OPERATORGRAPH formalism provides a
 66 template for decomposing forward models in other domains into typed, composable primi-
 67 tives with validated adjoints. The Triad Decomposition—separating information capacity,
 68 noise, and model fidelity as independent failure modes—applies wherever an inverse prob-
 69 lem is solved computationally. The practical lesson is domain-general: before investing in
 70 a better solver, diagnose whether the model is the bottleneck.

The Finite Primitive Basis

A foundational question in computational imaging is whether the diversity of imaging forward models—from coded aperture cameras to MRI scanners to electron microscopes—can be captured by a small, fixed set of primitive operators. We answer affirmatively.

The OperatorGraph representation. Every imaging forward model is encoded as a typed directed acyclic graph (DAG) in which each node wraps a single primitive physical operator and edges define the data flow from source to detector. Every primitive implements both a `forward()` and an `adjoint()` method, with validated adjoint consistency ensuring $\langle H\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, H^\dagger \mathbf{y} \rangle$ to within numerical precision. Each edge carries tensor shape and dtype metadata, enabling static validation before execution. We call this formalism the OPERATORGRAPH intermediate representation (IR).

Eleven canonical primitives. The OPERATORGRAPH IR defines a library of exactly 11 canonical primitives $\mathcal{B} = \{P, M, \Pi, F, C, \Sigma, D, S, W, R, \Lambda\}$:

#	Primitive	Notation	Physical action
1	Propagate	$P(d, \lambda)$	Free-space wave propagation
2	Modulate	$M(\mathbf{m})$	Element-wise multiplication (mask, coil, absorption)
3	Project	$\Pi(\theta)$	Radon line-integral projection
4	Encode	$F(\mathbf{k})$	Fourier-domain encoding (k -space)
5	Convolve	$C(\mathbf{h})$	Spatial convolution (PSF)
6	Accumulate	Σ	Summation over spectral/temporal axis
7	Detect	$D(g, \eta)$	Detector response (5 canonical families)
8	Sample	$S(\Omega)$	Sub-sampling on index set Ω
9	Disperse	$W(\alpha, a)$	Wavelength-dependent spatial shift
10	Scatter	$R(\sigma, \Delta\varepsilon)$	Direction change and/or energy shift
11	Transform	$\Lambda(f, \theta)$	Pointwise nonlinear physics operation

The Detect nonlinearity η is restricted to five canonical families (linear-field: $\eta(x) = gx$; logarithmic; sigmoid; intensity-square-law: $\eta(x) = g|x|^2$; coherent-field), each with at most 2 scalar parameters. The Transform primitive Λ applies a pointwise nonlinear function within the physics chain (not at the detector) and is restricted to five families: exponential attenuation ($e^{-\alpha x}$, Beer–Lambert), logarithmic compression, phase wrapping ($\arg(e^{ix})$), polynomial ($\sum a_k x^k$, $d \leq 5$), and saturation. Both Detect and Transform are pointwise and parameter-limited, preventing either from becoming a universal approximator (see¹ for formal definitions and non-universality proofs).

Physics-stage mapping. Each factor of a forward model is classified into one of six physics-stage families and mapped to primitives from $\mathcal{B}$: propagation $\rightarrow \{P, C\}$; elastic interaction $\rightarrow \{M\}$; inelastic interaction (scattering) $\rightarrow \{R\}$; pointwise nonlinear physics $\rightarrow \{\Lambda\}$; encoding–projection $\rightarrow \{\Pi, F\}$; detection–readout $\rightarrow \{\Sigma, S, W, C, D\}$.

Fidelity levels. Forward models can be written at increasing physical fidelity: linear shift-invariant (simplest), linear shift-variant, nonlinear ray/wave-based, and full-wave or Monte Carlo. All 11 primitives apply uniformly across every fidelity level. Linear stages are composed from the first 10 primitives; nonlinear stages (beam hardening, phase wrapping, saturation) are captured by Transform Λ . The theorem therefore covers the full fidelity spectrum without restriction.

Five physical carriers. The library spans five carrier families—photons, electrons, spins, acoustic waves, and particles—with 26 registered modality templates (12 with full end-to-end correction validation; Extended Data Table 2; Supplementary Table S3).

Theorem statement.

Definition 1 (Imaging Operator Class). $\mathcal{C}_{\text{img}}$ consists of all imaging forward models expressible as a finite sequential-parallel composition of stages—each either linear with bounded operator norm ($\|H_k\| \leq B$) or pointwise nonlinear with bounded Lipschitz constant ($\text{Lip}(\Lambda_k) \leq L$)—with stage count $K \leq N_{\text{max}}$.

Definition 2 (ε -Approximate Representation). A typed DAG G with nodes $V \subseteq \mathcal{B}$ is an ε -approximate representation of $H \in \mathcal{C}_{\text{img}}$ if $\sup_{\|\mathbf{x}\| \leq 1} \|H(\mathbf{x}) - H_G(\mathbf{x})\| / (\|H(\mathbf{x})\| + \delta) \leq \varepsilon$, where $\delta > 0$ is a regularization constant, and $|V| \leq N_{\text{max}}$, $\text{depth}(G) \leq D_{\text{max}}$.

Theorem 3 (Finite Primitive Basis). For every $H \in \mathcal{C}_{\text{img}}$, there exists a typed DAG G with $V \subseteq \mathcal{B}$ that is an ε -approximate representation of H .

Proof sketch. By Definition 1, $H = H_K \circ \dots \circ H_1$ with $K \leq N_{\text{max}}$. Each factor is classified into one of six physics-stage families and realized by primitives: propagation $\rightarrow P$ or C ; elastic interaction $\rightarrow M$; inelastic interaction $\rightarrow R$; pointwise nonlinear physics $\rightarrow \Lambda$; encoding–projection $\rightarrow \Pi$ or F ; detection–readout $\rightarrow$ finite composition from $\{\Sigma, S, W, C, D\}$. For linear stages, per-factor errors compose via sub-multiplicativity: $\|H - H_G\| \leq K \cdot \max_k(\varepsilon_k) \cdot B^{K-1}$. For nonlinear stages, the Lipschitz composition bound $\|H(\mathbf{x}) - \hat{H}(\mathbf{x})\| \leq \sum_k \varepsilon_k \prod_{j>k} \gamma_j$ applies, where γ_j is the Lipschitz constant of stage j . Both bounds yield $\leq \varepsilon$ under the stated regularity conditions. Full proof in Supplementary Note 12; formal primitive semantics and typed DAG denotation in¹. $\square$

Scope of $\mathcal{C}_{\text{img}}$. Covers: all clinical, scientific, and industrial imaging modalities—linear and nonlinear—including coded aperture, interferometric, projection, Fourier-encoded, acoustic, scattering, beam hardening, phase wrapping, multiple scattering, Monte Carlo systems, quantum state tomography, and relativistic systems. No modality is excluded. Formal definitions and proofs in¹.

Extension protocol. A new primitive is warranted when no DAG over $\mathcal{B}$ achieves $e_{\text{img}} \leq \varepsilon$ within the complexity bounds. The extension requires: (1) validated forward/adjoint,

(2) demonstrated representation gap, (3) error reduction below ε , (4) need by ≥ 2 modalities, (5) backward-compatible closure re-test. Scatter (R) was added via this protocol when Compton imaging produced $e_{\text{img}} = 0.34$ without it.

Held-out closure test. To validate Theorem 3 empirically, we conduct a closure test under a frozen protocol: the primitive library $\mathcal{B}$ (11 primitives), Detect families (5), Transform families (5), fidelity threshold ($\varepsilon = 0.01$), and complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$) are all frozen before evaluation. We test 5 held-out modalities (OCT, photoacoustic, SIM, phase-contrast X-ray, electron ptychography) and 3 exotic modalities (quantum ghost imaging, THz-TDS, Compton scatter imaging):

Modality	e_{img}	#Nodes/Depth	Triad transfer	New prim.?
OCT	< 0.01	5 / 3	Y	N
Photoacoustic	< 0.01	4 / 3	Y	N
SIM	< 0.01	4 / 3	Y	N
Phase-contrast X-ray	< 0.01	5 / 4	Y	N
Electron ptycho	< 0.01	4 / 3	Y	N
Ghost imaging	< 0.01	4 / 3	Y	N
THz-TDS	< 0.01	3 / 2	Y	N
Compton scatter	$< 0.01^*$	4 / 3	Y	Y (R)

*With R in library; $e_{\text{img}} > 0.01$ without R .

Max observed: 5 nodes / depth 4; median: 4 nodes / depth 3 (cf. bounds $N_{\text{max}} = 20$, $D_{\text{max}} = 10$).

Seven of eight modalities decompose with the frozen library. Quantum ghost imaging is operator-equivalent to a single-pixel camera at the image-formation level—sharing the canonical DAG $\text{Source} \rightarrow M \rightarrow \Sigma \rightarrow D$ despite fundamentally different source physics. THz-TDS requires only the coherent-field Detect family. Compton scatter imaging triggered the extension protocol (Section “Extension protocol” above): its representation gap ($e_{\text{img}} = 0.34 \gg \varepsilon$) met the falsifiable criterion for basis incompleteness, and the disciplined resolution—adding Scatter (R)—covers 5+ additional modalities (Raman, fluorescence, DOT, Brillouin) while preserving all existing decompositions. The basis grows from 9 to 11 (22% increase) while modality coverage grows by 19%+.

Basis-growth saturation. Plotting the number of distinct primitives K against the number of registered modalities N reveals clear saturation (Figure 7): 9 of 11 primitives are introduced by the first 10 modalities, Disperse (W) by CASSI-type spectral systems, Scatter (R) when Compton/Raman-class modalities enter, and Transform (Λ) when nonlinear physics stages (beam hardening, phase wrapping) are encountered. The growth is sublinear and saturating: $K = 11$ at $N = 30$, with no new primitive required for the most recent 18 modalities added. This saturation is consistent with Theorem 3: once all six physics-stage families are covered by primitives, new modalities compose existing primitives rather than requiring new ones.

Theorem tightness and minimality. The theoretical complexity bounds ($N_{\max} = 20$, $D_{\max} = 10$) are conservative: empirically, all 26 registered and 8 held-out modalities (~ 30 unique, accounting for 4 overlaps) require ≤ 6 nodes and depth ≤ 5 (held-out closure test table above), with the median modality using 4 nodes at depth 3. The gap between empirical and theoretical bounds reflects the compactness of real physical imaging chains. Moreover, the basis is *minimal*: for each of the 11 primitives, we exhibit a witness modality whose representation error exceeds ε when that primitive is removed (Supplementary Note 10, Proposition 1 therein; beam hardening CT witnesses the necessity of Transform Λ). Eight primitives ($P, M, \Pi, F, \Sigma, D, R, \Lambda$) are strictly necessary; the remaining three (C, S, W) are necessary under the stated complexity bounds.

The Triad Decomposition

The TRIAD DECOMPOSITION asserts that every failure in computational image recovery within $\mathcal{C}_{\text{img}}$ can be attributed to one or more of exactly three root causes, which we term *gates*. The three gates are mutually exclusive in their physical origin yet may co-occur and interact in any given measurement scenario.

Why exactly three gates. The reconstruction pipeline has exactly three inputs: a sensing geometry H that determines what information is captured (Gate 1), a physical carrier whose statistics determine the noise floor (Gate 2), and a forward model H_{nom} that the solver inverts (Gate 3). Any reconstruction error must trace to one of these inputs: information that was never captured (null space of H), information that was captured but corrupted (carrier noise), or information that was captured but misinterpreted (model error $H_{\text{nom}} \neq H_{\text{true}}$). This trichotomy is exhaustive: the MSE decomposes as $\text{MSE} \leq \text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$ (Supplementary Note 1), with no residual term. Solver suboptimality (e.g., insufficient iterations or regularization mismatch) is subsumed by Gate 3 in the Triad formalism, because the solver’s implicit forward model includes its regularization assumptions.

Gate 1: Recoverability. **Gate 1** asks whether the measurement encodes sufficient information about the signal of interest. Formally, if the forward operator $H \in \mathbb{R}^{m \times n}$ maps the unknown signal $\mathbf{x} \in \mathbb{R}^n$ to the measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$, then the null space $\mathcal{N}(H)$ defines signal components that are fundamentally invisible to the sensor. When $\mathcal{N}(H)$ is large, no solver can recover the missing information. **Gate 1** failures are intrinsic to the measurement design and can only be remedied by acquiring additional data.

Gate 2: Carrier Budget. **Gate 2** asks whether the signal-to-noise ratio (SNR) is sufficient. Every physical carrier—photons, electrons, spins, acoustic waves, particles—is subject to fundamental noise limits: shot noise for photon-counting systems, thermal noise

in electronic detectors, T_1/T_2 relaxation noise in magnetic resonance. When the carrier budget is too low, the measurement is dominated by noise and the reconstruction degrades regardless of operator fidelity.

Gate 3: Operator Mismatch. **Gate 3** asks whether the forward model assumed by the solver matches the true physics. The solver operates with a nominal operator H_{nom} , but the data were generated by a true operator H_{true} . When $H_{\text{nom}} \neq H_{\text{true}}$, the reconstruction targets a phantom inverse problem. **Gate 3** failures are insidious because they produce structured artifacts that mimic signal content, leading practitioners to blame the solver rather than the model. Sources include geometric misalignment (mask shift, rotation), parameter drift (coil sensitivity variation, gain instability), and model simplification.

Definition 4 (Triad Decomposition). *Let $H \in \mathcal{C}_{\text{img}}$ with measurement $\mathbf{y} = H\mathbf{x} + \mathbf{n}$ and solver using nominal operator H_{nom} . The reconstruction error decomposes as $\text{MSE} \leq \text{MSE}^{(G1)} + \text{MSE}^{(G2)} + \text{MSE}^{(G3)}$, where $\text{MSE}^{(G1)}$ is the null-space loss (Supplementary Eq. S1), $\text{MSE}^{(G2)}$ is the noise-floor term (Supplementary Eq. S3), and $\text{MSE}^{(G3)}$ is the mismatch-induced term (Supplementary Eq. S5).*

Theorem 5 (Gate 3 Dominance). *For any $H \in \mathcal{C}_{\text{img}}$ operating above its Gate 1 floor ($\gamma > \gamma_{\text{min}}$) and Gate 2 floor ($\text{SNR} > \text{SNR}_{\text{min}}$), Gate 3 is the binding constraint whenever $\|\delta\theta\|_{\mathbf{J}^\top \mathbf{J}} > (\sigma_n / \|\mathbf{x}\|_\infty) \cdot \kappa(\mathbf{H})$.*

Proof. See Supplementary Note 1, Proposition 2. □

Key finding: Gate 3 dominates. Across all 9 validated configurations (7 modalities), **Gate 3** is the dominant failure gate ($C_{\text{mismatch}} > \max(C_{\text{noise}}, C_{\text{recover}})$ in 9/9 cases; Supplementary Tables S12–S13 confirm that Gates 1–2 bind only at extreme compression or photon starvation). In CASSI, a sub-pixel mask shift degrades MST-L by 13.98 ± 0.62 dB (mean $\pm$ s.d. over 10 scenes); in MRI, a 5% coil sensitivity mismatch under clinically realistic multi-coil conditions (8 coils, $4\times$ acceleration) degrades reconstruction by 1.75–7.14 dB (Supplementary Note 11). Calibration—not solver design—is the underinvested bottleneck: a single forward-model correction recovers more reconstruction quality than the gap between traditional and state-of-the-art solvers.

Relationship to the Finite Primitive Basis. The two contributions are complementary: the Finite Primitive Basis provides a universal representation (every forward model is a DAG over 11 primitives¹), and the Triad provides a universal diagnostic law over that representation. The DAG structure makes Gate 3 diagnosis *actionable*: the MismatchAgent localizes the offending primitive node and corrects its parameters.

Consequences: Diagnosis and Correction

The two theoretical results directly imply a practical framework. The OPERATORGRAPH provides a modality-agnostic representation; the Triad provides root-cause diagnosis. Three deterministic agents—one per gate—evaluate information capacity, carrier budget, and operator mismatch respectively, producing a quantitative TRIADREPORT for every imaging configuration (see Methods for implementation details).

Correction pipeline. When **Gate 3** is dominant, a two-stage correction pipeline refines the forward model parameters: a coarse beam search over the mismatch parameter family $\theta = (\theta_1, \dots, \theta_k)$ associated with the offending DAG node, followed by gradient refinement via backpropagation through the differentiable forward model. Critically, correction operates exclusively on the forward model parameters, not on the solver weights: any existing solver—iterative, plug-and-play, or deep unrolling—benefits from the corrected operator without modification or retraining.

4-Scenario Protocol. To rigorously evaluate correction quality, PWM defines four canonical scenarios. **Scenario I** (Ideal): the solver reconstructs using the true operator H_{true} . **Scenario II** (Mismatch): the solver uses the nominal operator H_{nom} on data generated by H_{true} . **Scenario III** (Corrected): the solver uses the PWM-corrected operator. **Scenario IV** (Oracle): the true operator on mismatched data, providing the correction ceiling. The recovery ratio $\rho = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ quantifies how much of the mismatch-induced degradation is recovered (see Methods, Equation (5)).

Empirical Validation

We validate both theoretical contributions in two stages: controlled simulation experiments across twelve modalities using the 4-Scenario Protocol, and hardware validation on real data from ten modalities spanning all five carrier families (CASSI, CACTI, CT, ptychography, MRI, ultrasound, cryo-EM, CT detector offset, compressive holography, fluorescence microscopy).

Simulation experiments

Correction results. Extended Data Table 1 (Supplementary Table S1) summarizes the oracle correction ceiling across 14 configurations spanning 12 modalities. The oracle gain $\Delta_{\text{oracle}} = \text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}}$ ranges from $+0.76 \pm 0.12$ dB (CASSI/GAP-TV, 95% bootstrap CI over 10 scenes) to $+10.68 \pm 0.41$ dB (CT) across the original photon/X-ray modalities. Autonomous grid-search calibration achieves 82–100% of the oracle ceiling (Supplementary Table S9). The validated modalities span photon-domain (CASSI: $+0.76/+6.50$ dB [GAP-

TV/MST-L]; CACTI: $+10.21 \pm 0.85$ dB; SPC: $+7.71/+10.38$ dB [FISTA-TV/HATNet]; Lens-
less: $+3.55 \pm 0.22$ dB; compressive holography: $+0.0/+0.4$ dB; fluorescence microscopy:
 $+0.1/+6.8$ dB), coherent-photon (Ptychography: $+7.09 \pm 0.47$ dB), X-ray (CT: $+10.68 \pm$
 0.41 dB; CT offset: $+2.5/+3.9$ dB), acoustic (Ultrasound: $+0.1/+0.9$ dB), and electron
(Cryo-EM: $+0.1/+3.6$ dB) and nuclear spin (MRI: $+1.75$ to $+7.14$ dB under clinically re-
alistic multi-coil conditions at 3–5% sensitivity mismatch; Supplementary Note 11). All
confidence intervals are computed via the bootstrap percentile method with $B = 1,000$
resamples over test scenes (Methods). Correction gains are commensurate across all five
carrier families, confirming carrier-agnostic operation.

Modality deep dives. In CASSI, a 5-parameter mismatch¹⁷ collapses all solvers to
 ~ 21 dB regardless of ideal performance (24–35 dB); oracle recovery varies by solver ($\rho_{IV} = 0-$
 0.57 ; Supplementary Table S2), confirming that multi-parameter mismatches are harder to
correct than isolated shifts. In CACTI, EfficientSCI¹⁸ drops by 20.58 dB, yet GAP-TV
achieves 100% autonomous recovery of the oracle ceiling (Supplementary Table S9)—the
best ideal-condition solver suffers the largest degradation. SPC gain drift is corrected to
86–92% of the oracle ceiling (Table S9). **Gate 3** is dominant in every case; Gates 1–2 impose
irrecoverable information-theoretic limits only at extreme compression or photon starvation
(Supplementary Tables S12–S13).

Zero-shot generalization. Hyperparameters tuned on photon-domain modalities (CASSI,
CACTI, SPC) transfer without modification to coherent-photon, spin, and X-ray domains,
with comparable correction gains (Figure 4), confirming carrier-agnostic operation enabled
by the OperatorGraph abstraction.

Hardware validation

CASSI real data. These experiments use physical measurements from hardware-calibrated
coded aperture systems—not synthetic data—providing direct evidence that mismatch dom-
inance persists under real manufacturing tolerances and environmental conditions. We re-
construct 5 real TSA scenes¹⁰ under calibrated and perturbed masks. GAP-TV shows a
mean residual ratio of $1.8\times$ ($0.00189 \rightarrow 0.00333$); HDNet shows $1.0\times$ (mask-oblivious).
MST-S/L show ratios near $1.0\times$ on real data, in contrast to severe synthetic degrada-
tion. This asymmetry is itself informative: real hardware already contains unmodeled
manufacturing imperfections, and an additional software-simulated perturbation is par-
tially absorbed by pre-existing calibration errors (Supplementary Table S7). An extended
cross-residual analysis (Supplementary Note 15) reveals a deeper diagnostic: the *cross-*
residual—evaluating a mismatched reconstruction under the true forward model—increases
monotonically from 0.3% at 0.25 px shift to 11.1% at 2.0 px shift, with per-scene standard
deviation $< 0.4\%$, confirming that Gate 3 sensitivity is scene-independent and monotonic

in mismatch magnitude. These real-data results independently confirm the Triad predictions from simulation: the InverseNet benchmark¹⁷ validates the same mismatch-dominance pattern on physical DD-CASSI and CACTI instruments.

CACTI real data. GAP-TV shows $10.4\times$ mean residual ratio under sub-pixel shift, with per-scene ratios from $9.4\times$ to $11.0\times$. An extended analysis (Supplementary Note 15) reveals a striking dissociation: the *self-residual* barely changes (1.0 – $1.12\times$) because the solver adapts to the wrong model, but the *cross-residual* explodes 44 – $462\times$ as the shift increases from 0.25 to 1.0 px. This demonstrates that self-consistency is not a reliable diagnostic for operator mismatch in underdetermined systems, with direct implications for compressive sensing quality control.

CT real sinograms. We extend hardware validation beyond optical instruments to three additional carrier families. For **X-ray CT**, two public sinogram datasets—the FIPS walnut micro-CT (1200 projections, 2296 detectors) and the Helsinki Tomography Challenge 2022 (721 projections, 560 detectors)—show that center-of-rotation (CoR) offsets of 1–8 pixels cause monotonic PSNR degradation of 9.4 dB (walnut) and 8.0 dB (HTC), with 100% oracle recovery (Supplementary Note 15). For **electron ptychography**, real 4D-STEM data from SrTiO₃ (Zenodo 5113449; 300 kV, 128×128 scan) shows that probe position jitter of 0.5–4.0 pixels causes monotonic iCoM phase degradation from 35.5 to 19.4 dB (16.1 dB total loss), with $> 99.9\%$ oracle recovery (Supplementary Note 15). For **MRI**, multi-coil brain k -space from M4Raw (Zenodo 8056074; 4 coils, 256×256) shows that coil sensitivity perturbation up to 40% degrades R=2 SENSE reconstruction by 0.5 dB, with 100% recovery via sensitivity re-estimation. The modest MRI effect is consistent with the over-determined encoding at R=2; the simulation experiments at R=4 (Supplementary Note 11) confirm that Gate 3 severity scales with acceleration factor.

Ultrasound. The acoustic carrier family—absent from all prior validation—is tested via simulated pulse-echo B-mode imaging with a 64-element linear array at 5 MHz centre frequency and 100 MHz sampling rate on a 128×128 tissue phantom. Speed-of-sound (SoS) mismatch of $[50, 100, 200, 400]$ m/s (relative to true $c = 1540$ m/s) causes monotonic PSNR degradation from 0.1 to 0.9 dB, with delay-and-sum (DAS) beamforming as the reconstruction solver. Grid-search calibration over 21 SoS candidates in $[1440, 1640]$ m/s achieves $\rho = 1.19$ – 2.77 (Supplementary Table S14); the over-unity ratios indicate that the optimizer consistently finds an operating point ($c = 1440$ m/s) that improves upon oracle reconstruction, reflecting the DAS beamformer’s inherent model simplifications. This confirms Gate 3 dominance for acoustic propagation, completing the five-carrier-family validation.

Cryo-EM. Electron-carrier validation uses simulated cryo-EM micrographs (128×128 , pixel size 0.1 nm) with contrast transfer function (CTF) encoding. Defocus mismatch of

[100, 200, 500, 1000] nm (relative to true $\Delta f = -2000$ nm) degrades Wiener-filter reconstruction by 0.1–3.6 dB. At Nyquist frequency 5 nm^{-1} , the CTF exhibits ~ 30 oscillations before the B-factor envelope ($B = 2 \text{ nm}^2$) attenuates high frequencies, making the reconstruction highly sensitive to defocus error. Grid-search calibration over 51 defocus values in $[-3000, -500] \text{ nm}$ recovers $\rho = 1.00$ – 1.12 (Supplementary Table S15), confirming that CTF parameter mismatch is the dominant bottleneck in single-particle reconstruction—consistent with the extensive defocus-estimation literature in structural biology.

CT detector offset. X-ray CT validation uses a 128×128 Shepp–Logan phantom with 360-angle parallel-beam projections. Detector offset mismatch of [2, 5, 10, 20] pixels—simulated by shifting the sinogram along the detector axis—causes 2.5–3.9 dB PSNR degradation under filtered backprojection (FBP) with Shepp–Logan filter. At large offsets (≥ 10 px), edge data loss additionally reduces the oracle PSNR from 13.8 to 10.6 dB (a Gate 1 effect). Grid-search calibration over 21 offset candidates in $[-25, 25]$ px achieves $\rho = 0.83$ – 1.00 (Supplementary Table S16), with exact offset recovery (≤ 0.5 px error) for offsets ≥ 5 pixels.

Compressive holography. Multi-depth Fresnel propagation encodes a 3D object ($4 \times 64 \times 64$) into a single 2D hologram—the most compressive encoding among validated modalities ($\lambda = 532 \text{ nm}$, pixel pitch $5 \mu\text{m}$, depth spacing $200 \mu\text{m}$). Propagation distance error of [10, 50, 100, 200] μm causes progressive defocus, degrading FISTA-TV reconstruction by up to 0.4 dB at $200 \mu\text{m}$ error. Autonomous calibration via hologram residual minimisation ($\|y - A_{\text{cal}}\hat{x}\|^2/\|y\|^2$) achieves $\rho = 0.64$ – 1.59 (Supplementary Table S17). The per-plane PSNR analysis reveals that deeper planes are more sensitive to distance error, consistent with the quadratic phase scaling of Fresnel kernels.

Fluorescence microscopy. Dual-PSF Stokes-shift imaging—excitation and emission wavelengths produce distinct PSF widths—is validated under PSF sigma mismatch of [0.1, 0.3, 0.5, 1.0] pixels on a 64×64 fluorescence phantom (1000 peak photons). Richardson–Lucy deconvolution (80 iterations) degrades by 0.1–6.8 dB. A 2D grid search over $(\sigma_{\text{ex}}, \sigma_{\text{em}})$ with measurement-residual minimisation achieves $\rho = 0.44$ – 0.75 (Supplementary Table S18). Recovery is lowest at the smallest error ($\Delta\sigma = 0.1$ px, $\rho = 0.44$) where the calibration signal is near the noise floor, and strongest at moderate errors ($\Delta\sigma = 0.5$ px, $\rho = 0.75$) where the objective landscape is well-resolved.

Autonomous calibration. Grid-search calibration recovers $\rho = 0.85$ (CASSI, 1,140 s; 95% CI [0.79, 0.91]), $\rho = 1.00$ (CACTI, 60 s; CI [0.97, 1.00]), $\rho = 0.86$ – 0.92 (SPC via TV objective; CI width ≤ 0.06), $\rho = 1.19$ – 2.77 (Ultrasound SoS), $\rho = 1.00$ – 1.12 (Cryo-EM defocus), $\rho = 0.83$ – 1.00 (CT detector offset), $\rho = 0.64$ – 1.59 (compressive holography), and $\rho = 0.44$ – 0.75 (fluorescence PSF) of the oracle correction. Recovery ratios exceeding unity (ultrasound, cryo-EM, compressive holography) indicate that the optimizer found

operating points marginally superior to the assumed ground-truth parameters, reflecting inherent model simplifications in the reconstruction algorithms. Single-parameter modalities (CACTI, CT) achieve near-perfect or above-unity recovery because their mismatch manifolds are low-dimensional with clean objective landscapes. CT CoR calibration on real sinograms also achieves $\rho = 1.00$ (CI [0.99, 1.00]). Multi-parameter modalities (CASSI, fluorescence) show lower but still substantial recovery, reflecting the higher-dimensional search space.

Simulation-to-hardware gap. CASSI shows smaller real degradation ($1.8\times$) than predicted by simulation, because as-built masks already contain unmodeled imperfections that absorb incremental perturbations. CACTI shows larger real sensitivity ($10.4\times$) than simulation, because the temporal mask pattern replicates errors across all compressed frames. This bidirectional discrepancy—invisible without a unified cross-modality framework—carries a methodological lesson: synthetic mismatch studies can both overestimate marginal impact (CASSI) and underestimate cumulative sensitivity (CACTI).

Discussion

The central finding is that the space of imaging forward models has finite structural complexity—11 primitives suffice and are necessary—and that operator mismatch, not information deficiency or noise, is the dominant reconstruction bottleneck across all validated modalities. The implication is not that solver research is unnecessary, but that calibration is systematically underinvested: a single forward-model correction step often recovers more reconstruction quality than the gap between a classical solver and a state-of-the-art deep network operating on the same mismatched operator.

Implications for autonomous scientific discovery. The bounded complexity of imaging physics has consequences that extend beyond calibration. If every imaging forward model—from a \$10 webcam to a particle accelerator beamline—decomposes into the same 11 primitives, then any autonomous system tasked with “learning to see” faces a *finite search problem* over typed physical operators, not an open-ended function approximation. This re-frames the challenge for AI Scientists¹⁹: rather than discovering physical laws from scratch, a machine learning system equipped with the OPERATORGRAPH representation needs only to identify which subset of 11 known primitives is active in a new instrument and estimate their parameters from data. The Triad Decomposition further constrains the hypothesis space for failure: an AI Scientist confronting a degraded reconstruction has exactly three root causes to consider, each with a formal test and a known correction strategy. The observation that 9 of 11 primitives are introduced by the first 10 modalities, with the final two (Disperse and Scatter) appearing only for exotic carrier physics, suggests that the primitive

manifold is dense at its core and sparse at its boundary—a structure that should enable few-shot generalization to new modalities from existing primitive combinations. More broadly, the Finite Primitive Basis Theorem raises a question for computational science beyond imaging: if the forward model of every physical measurement system can be expressed in a small universal grammar, what analogous grammars exist for other physical processes, and what does their existence imply for the long-term trajectory of physics-informed machine learning?

Comparison with specialist calibration. Specialist calibration methods—ESPIRiT⁸ for MRI, ePIE⁹ for ptychography, cross-correlation auto-focus for CT, CTFFIND for cryo-EM—outperform PWM on their home modality when adequate calibration data is available (Supplementary Note S14). However, they degrade under data-limited conditions (ESPIRiT: −9.29 dB at 24 ACS lines), require per-modality engineering, and do not exist for several validated modalities (CASSI, CACTI, compressive holography). PWM’s value is *generality*: a single pipeline serving 12 modalities across all five carrier families with robust performance across calibration data regimes. The two approaches are complementary—specialist estimates can initialize PWM correction, and PWM can refine specialist estimates.

Hardware validation interpretation. The hardware results reveal that Gate 3 sensitivity depends on the *condition number* of the encoding matrix. On real CASSI, perturbation produces smaller degradation (1.8×) than predicted, because the as-built mask already contains unmodeled manufacturing imperfections. On real CACTI, degradation is severe (10.4×), because the simpler temporal-mask optics lack this pre-existing error buffer. On real CT sinograms, CoR offset causes 8–9 dB loss; electron ptychography shows up to 16.1 dB phase degradation. This range—from sub-dB (well-conditioned MRI at R=2) to > 16 dB (ptychography)—is itself a Triad prediction. The extended cross-residual analysis further reveals that *self-consistency metrics are misleading* in underdetermined systems (CACTI cross-residual is 462× while self-residual is only 1.12×), underscoring the need for forward-model-aware diagnostics.

Boundary conditions and failure modes. The framework has well-defined limitations that inform its scope of applicability:

- **Multi-parameter mismatch:** CASSI recovery ratios under 5-parameter mismatch are moderate ($\rho = 22\text{--}46\%$), reflecting the inherent difficulty of simultaneously correcting multiple coupled parameters on a high-dimensional manifold. Single-parameter correction achieves 85–100%.
- **Nonlinear forward models:** The Triad diagnostic has been validated on linear forward models only; beam hardening CT and phase-wrapped MRI require additional

validation (the 11-primitive basis formally covers these via Transform Λ^1).

- **Undeclared mismatch:** Correction is limited to declared mismatch parameter families in the mismatch database; undeclared failure modes (e.g., nonlinear detector drift) require extending the database.
- **Software-perturbation protocol:** Current real-data experiments apply software-simulated perturbations to existing measurements rather than physically displacing hardware and re-acquiring; the controlled hardware protocol is specified in Methods and Supplementary Note 8.

Falsifiable predictions. The framework generates concrete, testable predictions:

1. **SIM:** Gate 3 should dominate when pattern phase error exceeds 0.05 rad. Predicted correction: +3–8 dB.
2. **OCT:** Dispersion mismatch should produce +2–5 dB correction gain via the Disperse primitive (W).
3. **Electron ptychography:** Position errors exceeding 1/10 of the probe diameter should trigger Gate 3 dominance. This prediction is *confirmed* by real 4D-STEM validation: +5 to +16 dB correction on SrTiO₃ data (Supplementary Note 15).

These predictions are falsifiable: if any modality shows Gate 1 or Gate 2 dominance under the stated conditions, the framework would require revision. Hardware-in-the-loop validation with physical parameter displacement is the natural next step (Supplementary Note 8).

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Author Contributions. C.Y. conceived the project, designed the TRIAD DECOMPOSITION framework, proved the Finite Primitive Basis Theorem, developed the OPERATOR-GRAPH IR, implemented the agent and correction systems, performed all simulation and real-data experiments, and wrote the manuscript. X.Y. developed the GAP-TV reconstruction algorithm used as the primary solver across CASSI, CACTI, and SPC experiments, contributed the EfficientSCI architecture used for CACTI validation, provided the CASSI and CACTI forward model specifications and mismatch parameter characterizations that define the 5-parameter mismatch model, validated the real-data experimental protocols for both CASSI (TSA scenes) and CACTI instruments, and edited the manuscript.

Competing Interests. C.Y. is an employee of NextGen PlatformAI C Corp, which develops the PWM platform. The authors declare no other competing interests.

467 **Ethics Declarations.** This study does not involve human participants, human tissue, or
468 animals. All experiments use publicly available benchmark datasets and simulated data.

469 **Data Availability.** All synthetic measurement data can be regenerated using the OPER-
470 ATORGRAPH templates and mismatch parameters in the Supplementary Information. The
471 KAIST hyperspectral dataset²⁰ and TSA real-data scenes used for CASSI experiments are
472 publicly available. CACTI real-data scenes are available from the EfficientSCI repository¹⁸.
473 CT sinograms are from the FIPS walnut dataset (Zenodo 1254206) and Helsinki Tomogra-
474 phy Challenge 2022 (Zenodo 6984868). The 4D-STEM ptychography dataset (SrTiO₃ [001])
475 is from Zenodo 5113449. Multi-coil MRI k -space data (M4Raw) is from Zenodo 8056074.

476 **Code Availability.** The PWM codebase, including all OPERATORGRAPH templates, agent
477 implementations, real-data validation scripts, and evaluation pipelines, is available at https://github.com/integritynoble/Physics_World_Model
478 under the PWM Noncommercial
479 Share-Alike License v1.0 (see LICENSE in the repository).

480 **Correspondence.** Correspondence and requests for materials should be addressed to
481 C.Y. (integrityyang@gmail.com).

482 Online Methods

483 OperatorGraph Specification

484 **Formal definition.** The OPERATORGRAPH intermediate representation encodes the for-
485 ward physics of any computational imaging modality as a directed acyclic graph (DAG)
486 $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Each node $v_i \in \mathcal{V}$ wraps a *primitive operator* and implements two entry points:
487 `forward( $x$ )  $\rightarrow$   $y$` and `adjoint( $y$ )  $\rightarrow$   $x$` , the latter defined only when the primitive is lin-
488 ear. Edges $e_{ij} \in \mathcal{E}$ encode data flow: the output of node v_i is passed to node v_j . Each
489 node additionally exposes a set of learnable parameters θ_i that may be perturbed during
490 mismatch simulation or optimized during calibration, as well as read-only metadata flags
491 (`is_linear`, `is_stochastic`, `is_differentiable`). The graph is stored as a declarative
492 YAML specification (`OperatorGraphSpec`) and compiled to an executable `GraphOperator`
493 object by the `GraphCompiler`.

494 **Node types.** Primitive operators fall into two categories:

- 495 • **Linear operators.** Convolution (`conv2d`), mask modulation (`mask_modulate`), sub-
496 pixel shift (`subpixel_shift_2d`), Radon transform (`radon_fanbeam`), Fourier encod-
497 ing (`fourier_encode`), spectral dispersion (`spectral_disperse`), Fresnel propagation
498 (`fresnel_propagate`), random projection (`random_project`), and structured illumi-
499 nation (`sim_modulate`). Each implements both `forward()` and `adjoint()`.

500 • **Nonlinear operators.** Squared magnitude (`magnitude_sq`), Poisson–Gaussian noise
 501 (`poisson_gaussian`), saturation clipping (`saturation_clip`), phase retrieval nonlin-
 502 earity (`phase_abs`), and detector quantization (`quantize`). These set `is_linear` =
 503 `False` and raise `NotImplementedError` on `adjoint()`, except where a well-defined
 504 pseudo-adjoint exists (*e.g.*, the identity adjoint for magnitude-squared in Gerchberg–
 505 Saxton-type algorithms).

506 **Adjoint consistency and Lipschitz bounds as mathematical safety rails.** Two
 507 structural constraints distinguish the OPERATORGRAPH IR from standard deep-learning
 508 forward models and prevent either nonlinear primitive family from becoming a universal
 509 approximator.

510 First, *adjoint consistency*: correctness of every linear primitive is verified by a ran-
 511 domized dot-product test. For a primitive A with forward map $A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, we draw
 512 $x \sim \mathcal{N}(0, I_n)$ and $y \sim \mathcal{N}(0, I_m)$ and compute

$$\delta = \frac{|\langle A^*y, x \rangle - \langle y, Ax \rangle|}{\max(|\langle A^*y, x \rangle|, \epsilon)} \quad (1)$$

513 where $\epsilon = 10^{-12}$ guards against division by zero. The test is repeated $n_{\text{trials}} = 5$ times
 514 with independent random draws; the primitive passes if $\delta_{\text{max}} < 10^{-6}$. At the graph level,
 515 a compiled `GraphOperator` composed entirely of linear nodes executes the same test over
 516 the composed forward–adjoint chain. A `GraphAdjointCheckReport` records n_{trials} , δ_{max} ,
 517 and $\bar{\delta}$ for audit. All graph templates that consist solely of linear primitives pass this check.
 518 Adjoint consistency is not merely a numerical correctness guarantee: it is a *physical faithful-*
 519 *ness certificate*. A model whose adjoint is inconsistent implements a physically impossible
 520 energy accounting, making gradient-based calibration unreliable and certifiable reconstruc-
 521 tion bounds vacuous. Neural network forward models lack this certificate by construction,
 522 since their weight matrices have no physical meaning and their transposes are not computed
 523 or validated.

524 Second, *Lipschitz bounds*: the two nonlinear primitive families, Detect D and Trans-
 525 form Λ , are each restricted to five canonical function families with at most two scalar param-
 526 eters. This restriction enforces bounded Lipschitz constants ($\text{Lip}(\Lambda_k) \leq L$ for the explicit L
 527 derived in¹) and prevents either primitive from becoming a universal approximator in the
 528 sense of Cybenko²¹. Concretely: a generic neural network layer with sigmoid activations
 529 can approximate any continuous function on a compact domain; neither Detect nor Trans-
 530 form can, because their function families are closed under composition only within the five
 531 enumerated types. This non-universality is the key property that makes the 11-primitive li-
 532 brary *falsifiable*: if a modality’s forward physics cannot be represented within the restricted
 533 families, the extension protocol (Main Text, “Extension protocol”) fires and a new primitive
 534 must be justified physically. The combination of adjoint consistency and bounded Lipschitz

535 nonlinearity provides the mathematical safety rails that make OPERATORGRAPH-based re-
 536 construction both interpretable and certifiable—properties that standard physics-informed
 537 neural networks²² achieve only approximately and without modality-agnostic guarantees.

538 **Graph compilation.** The compiler executes a four-stage pipeline:

- 539 1. **Validate.** Confirm acyclicity via topological sort (Kahn’s algorithm), verify that ev-
 540 ery `primitive_id` exists in the global `PRIMITIVE_REGISTRY`, reject duplicate `node_id`
 541 values, and optionally verify shape compatibility along edges when a `canonical_chain`
 542 metadata flag is set.
- 543 2. **Bind.** Instantiate each primitive with its parameter dictionary θ_i .
- 544 3. **Plan forward.** The topological sort yields a sequential execution plan $(v_{\pi(1)}, \dots, v_{\pi(|V|)})$.
- 545 4. **Plan adjoint.** For graphs where `all_linear = True`, the adjoint plan reverses the
 546 topological order and applies each node’s individual adjoint in sequence, implementing
 547 the chain rule $A^* = A_1^* \circ \dots \circ A_{|V|}^*$ for a composition $A = A_{|V|} \circ \dots \circ A_1$. For
 548 graphs containing nonlinear nodes, the adjoint plan is not generated, and any call to
 549 `adjoint()` raises `NotImplementedError` at runtime.

550 The compiled `GraphOperator` is serializable to JSON and hashable via SHA-256 for prove-
 551 nance tracking in `RunBundle` manifests.

552 **Template library.** The `graph_templates.yaml` registry contains 90 templates spanning
 553 all five carrier families. Of these, 26 modalities carry formal validation status (7 with
 554 full end-to-end correction validation, 1 with Scenario I baseline, 18 with template-level
 555 validation; Supplementary Table S3). Representative modalities by carrier family include:

- 556 • **Photons (optical and X-ray):** CASSI, SPC, CACTI, structured illumination
 557 microscopy (SIM), confocal, light-sheet, holography, ptychography, Fourier ptycho-
 558 graphic microscopy (FPM), optical coherence tomography (OCT), lensless imaging,
 559 light field, integral imaging, neural radiance fields (NeRF), Gaussian splatting, fluo-
 560 rescence lifetime imaging (FLIM), diffuse optical tomography (DOT), phase retrieval,
 561 X-ray computed tomography (CT), and cone-beam CT (CBCT).
- 562 • **Electrons:** Electron diffraction, electron backscatter diffraction (EBSD), electron
 563 energy loss spectroscopy (EELS), and electron holography.
- 564 • **Spins (MRI):** Functional MRI (fMRI), diffusion-weighted MRI (DW-MRI), and
 565 magnetic resonance spectroscopy (MRS).
- 566 • **Acoustic:** Ultrasound B-mode, Doppler ultrasound, shear-wave elastography, sonar,
 567 and photoacoustic tomography (combines optical excitation with acoustic detection).
- 568 • **Particles:** Neutron tomography, proton radiography, and muon tomography.

569 **Fidelity levels.** Each template is parameterized by a fidelity level that controls the degree
570 of physical realism in the simulated forward model:

571 **Level 1 (Linear, shift-invariant):** The forward model is a linear, spatially uniform operator—
572 the simplest approximation, suitable for initial diagnostics and rapid prototyping.

573 **Level 2 (Linear, shift-variant):** Spatially varying operator parameters (e.g. non-uniform
574 illumination, position-dependent PSF, multi-coil sensitivity maps in MRI). Adds a
575 modality-appropriate noise model (Poisson shot noise plus Gaussian read noise for
576 photon-counting modalities, Rician noise for MRI, Poisson for CT).

577 **Level 3 (Nonlinear, ray/wave-based):** Includes nonlinear effects such as beam hard-
578 ening, phase wrapping, and scattering, captured by the Transform Λ primitive. Per-
579 turbation families and ranges are specified in `mismatch_db.yaml`.

580 **Level 4 (Full-wave / Monte Carlo):** Complete physical simulation including wave-optical
581 propagation, spatially varying aberrations, detector nonlinearities, and environmental
582 drift. Currently implemented for holography and ptychography.

583 All 11 primitives in the library $\mathcal{B}$ apply uniformly across every fidelity level; the levels
584 organize simulation complexity, not theorem scope.

585 Triad Decomposition Formalization

586 The TRIAD DECOMPOSITION asserts that the quality of any computational imaging re-
587 construction is bounded by three fundamental gates. Rather than a qualitative guideline,
588 PWM quantifies each gate numerically and uses the resulting scores to diagnose the domi-
589 nant bottleneck in any imaging configuration.

590 **Gate 1 (Recoverability).** Recoverability measures the information-theoretic capacity
591 of the sensing geometry. We quantify it via the *effective compression ratio* $r = m/n$, where
592 m is the number of independent measurements and n the dimension of the signal. The
593 `compression_db.yaml` registry (1,186 lines) stores, for each modality, a lookup table map-
594 ping compression ratio to expected reconstruction PSNR under ideal conditions, obtained
595 from calibration experiments or published benchmarks. Each entry carries a **provenance**
596 field citing the source (paper DOI, internal experiment ID, or theoretical formula). Addi-
597 tional recoverability indicators include the effective rank of the measurement matrix (esti-
598 mated via randomized SVD for large operators), the dimension of the null space, and the
599 restricted isometry property (RIP) constant where analytically tractable (*e.g.*, for Gaussian
600 random projections in SPC).

601 **Gate 2 (Carrier Budget).** The carrier budget quantifies the signal-to-noise ratio (SNR)
 602 of the measurement channel. The **PhotonAgent** consumes the **photon_db.yaml** registry
 603 (624 lines) which stores, per modality, a deterministic photon model parameterized by
 604 source power, quantum efficiency, exposure time, and detector characteristics. The agent
 605 classifies the noise regime into one of three categories: *shot-limited* (Poisson-dominated,
 606 $\text{SNR} \propto \sqrt{N_{\text{photon}}}$), *read-limited* (Gaussian read noise dominates, $\text{SNR} \propto N_{\text{photon}}/\sigma_{\text{read}}$),
 607 and *dark-current-limited* (long exposures where dark current accumulation dominates). The
 608 output is a **PhotonReport** containing the estimated SNR in decibels, the noise regime
 609 classification, per-element photon count, and a feasibility verdict (**sufficient**, **marginal**,
 610 or **insufficient**).

611 **Gate 3 (Operator Mismatch).** Operator mismatch quantifies the discrepancy between
 612 the assumed forward model H_{nom} and the true physical operator H_{true} . The **MismatchAgent**
 613 consults **mismatch_db.yaml** (797 lines) which catalogs, for each modality, the set of mis-
 614 match parameters (spatial shifts, rotational offsets, dispersion errors, PSF deviations, coil
 615 sensitivity errors, center-of-rotation offsets, *etc.*), their typical ranges, and available cor-
 616 rection methods. The mismatch severity score $s \in [0, 1]$ is computed as the normalized ℓ_2
 617 distance $\|\boldsymbol{\theta}_{\text{true}} - \boldsymbol{\theta}_{\text{nom}}\|/\|\boldsymbol{\theta}_{\text{range}}\|$, where $\boldsymbol{\theta}_{\text{range}}$ is the per-parameter dynamic range from the
 618 registry. Sensitivity analysis $\partial\text{PSNR}/\partial\theta_k$ is estimated via finite differences on the forward
 619 model. The output is a **MismatchReport** containing the severity score, the dominant mis-
 620 match parameter, the recommended correction method, and the expected PSNR gain from
 621 correction.

622 **Gate binding determination.** Given reconstruction results under the four-scenario pro-
 623 tocol (the Evaluation Protocol section below), PWM identifies the dominant gate by com-
 624 paring three cost terms:

$$C_{\text{mismatch}} = \text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}} \quad (2)$$

$$C_{\text{noise}} = \text{PSNR}_{\text{ideal}} - \text{PSNR}_{\text{noisy}} \quad (3)$$

$$C_{\text{recover}} = \text{PSNR}_{\text{limit}} - \text{PSNR}_{\text{I}} \quad (4)$$

625 where PSNR_{I} is the reconstruction PSNR under Scenario I (ideal operator), PSNR_{II} under
 626 Scenario II (mismatched operator), $\text{PSNR}_{\text{noisy}}$ under the corresponding noisy condition,
 627 and $\text{PSNR}_{\text{limit}}$ is the theoretical upper bound from the compression table. The dominant
 628 gate is $\arg \max_g C_g$.

629 **TriadReport.** For every diagnosis, PWM produces a **TRIADREPORT**: a Pydantic-validated
 630 structured artifact comprising **dominant_gate** (enum: **recoverability**, **carrier_budget**,
 631 **operator_mismatch**), **evidence_scores** (three floats, one per gate), **confidence_interval**

(float, 95% CI width from bootstrap), `recommended_action` (string, *e.g.* “increase compression ratio” or “apply mismatch correction”), and `parameter_sensitivities` (dictionary mapping each mismatch parameter name to its $\partial\text{PSNR}/\partial\theta_k$ value). The TRIADREPORT is mandatory—PWM does not permit a reconstruction to be reported without an accompanying diagnosis.

Recovery ratio. We define the *recovery ratio*

$$\rho = \frac{\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}}{\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}}} \quad (5)$$

which lies in $[0, 1]$ under standard convexity conditions (see Supplementary Note 1 for formal analysis; values $\rho > 1$ are possible when the corrected operator provides beneficial regularization). $\rho = 0$ indicates that calibration yields no benefit (mismatch is not the bottleneck), while $\rho = 1$ indicates that calibration fully closes the mismatch gap.

Agent System Architecture

The PWM agent system comprises 6 specialist agents, 1 optional hybrid agent, and 8 support classes totalling 10,545 lines of Python. All agents execute deterministically; no large language model (LLM) is required for pipeline operation.

PlanAgent. The orchestrator agent. Given a user prompt or a structured `ExperimentSpec`, PlanAgent parses the intent (`simulate`, `operator_correction`, or `auto`), maps the requested modality to its canonical key via the `modalities.yaml` registry (which contains 64 modality entries with keywords, forward model equations, and default solvers), builds an `ImagingSystem` contract, and dispatches to the appropriate sub-agents. When the mode is `auto`, PlanAgent inspects the available data and operator specification to determine whether simulation or operator correction is more appropriate.

PhotonAgent. Computes SNR feasibility deterministically from the `photon_db.yaml` registry. For each modality and photon-level tier (`bright`, `standard`, `low_light`), the agent evaluates the photon budget by combining source power, quantum efficiency, exposure time, and noise model parameters. The output `PhotonReport` is a strict Pydantic model containing `noise_regime` (enum), `snr_db` (float), `feasibility` (enum), and `per_element_photons` (float).

RecoverabilityAgent. A table-driven agent that consults `compression_db.yaml` (1,186 lines) to map the modality and compression ratio to an expected PSNR range. Each table entry includes provenance metadata citing the original source. The output `RecoverabilityReport` contains `compression_ratio`, `psnr_prediction`, `feasibility`, and `null_space_dim` where available.

664 **MismatchAgent.** Scores the mismatch severity for a given imaging configuration using `mismatch_db.yaml` (797 lines). For each modality, the database enumerates the relevant mismatch parameters, their physical units, typical perturbation ranges, and available correction algorithms. The output `MismatchReport` includes `severity` (float, 0–1),
666 `correction_method` (string), `expected_gain_db` (float), and `dominant_parameter` (string).
668

669 **AnalysisAgent.** The bottleneck classifier. It receives reports from the Photon, Recoverability, and Mismatch agents, computes the gate costs (Equations (2) to (4)), identifies the
670 dominant gate, and generates actionable suggestions. The `AnalysisAgent` also computes
671 the recovery ratio ρ and its bootstrap confidence interval.
672

673 **AgentNegotiator.** Implements a cross-agent veto protocol. Before reconstruction is authorized, the negotiator inspects all three upstream reports and applies three veto conditions: (1) low photon budget combined with aggressive compression (C_{noise} and C_{recover} both large); (2) severe mismatch (severity > 0.7) without a planned correction step; (3) joint
674 probability below the floor threshold ($p_{\text{joint}} < 0.15$), indicating that all three subsystems
675 are simultaneously marginal. When any veto fires, reconstruction halts with an actionable
676 explanation and suggested remediation.
677
678
679

680 **HybridAgent.** An optional wrapper that invokes an LLM for natural-language narrative generation or edge-case modality mapping. All quantitative decisions remain on the
681 deterministic code path; the `HybridAgent` is never required for pipeline operation.
682

683 **Support classes.** The remaining components include: `AssetManager` (file I/O and caching for large arrays), `ContinuityChecker` (verifies that sequential pipeline outputs are dimensionally consistent), `SystemDiscern` (auto-detects modality from uploaded data), `PreflightChecker`
684 (validates the complete experiment configuration before execution), `WhatIfPrecomputer`
685 (evaluates counterfactual what-if scenarios), `SelfImprovement` (logs diagnostic events for future registry refinement), `PhysicsStageVisualizer` (generates intermediate visualizations at each pipeline stage), and `UPWMI` (Universal Physics World Model Interface, the
686 top-level entry point that wires all agents together).
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691 **Contract system.** Inter-agent communication uses 25 Pydantic v2 contract models. All contracts inherit from `StrictBaseModel`, which enforces `extra="forbid"` (no unexpected fields), `validate_assignment=True` (mutations re-validated), and a model validator that
692 rejects NaN and Inf in any float field. Bounded scores use `Field(ge=0.0, le=1.0)`. Enums are string enums for human-readable JSON serialization. This design ensures that pipeline
693 failures surface immediately as validation errors rather than propagating silently.
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YAML registries. The system is driven by 9 YAML registries totalling 7,034 lines: `modalities.yaml` (modality definitions), `graph_templates.yaml` (OperatorGraph skeletons), `photon_db.yaml` (photon models), `mismatch_db.yaml` (mismatch parameters and correction methods), `compression_db.yaml` (recoverability tables with provenance), `solver_registry.yaml` (solver configurations), `primitives.yaml` (primitive operator metadata), `dataset_registry.yaml` (dataset locations and formats), and `acceptance_thresholds.yaml` (pass/fail thresholds per metric).

Correction Algorithms

We implement two complementary algorithms for operator mismatch correction. Crucially, both algorithms operate on the forward operator parameters θ rather than the reconstruction solver weights, making them *solver-agnostic*: the corrected operator $H(\hat{\theta})$ benefits any downstream solver (GAP-TV, MST-L, HDNet²³, CST, *etc.*) without retraining.

Algorithm 1: Hierarchical Beam Search. The coarse correction phase employs a hierarchical search strategy to rapidly explore the mismatch parameter space. For CASSI, the five-parameter mismatch model comprises mask affine parameters (spatial shifts dx , dy and rotation θ) and dispersion parameters (slope a_1 and axis angle α); an optional sixth parameter, PSF width σ_{psf} , is available but not used in the primary experiments. The algorithm proceeds as follows:

1. **1D sweeps.** Each parameter is swept independently over its full range while holding others at nominal values. This produces five 1D cost curves from which coarse optima are extracted.
2. **3D beam search.** The mask affine subspace (dx, dy, θ) is searched over a $5 \times 5 \times 5$ grid centered on the 1D optima. The top- k ($k = 5$) candidates by reconstruction PSNR are retained.
3. **2D beam search.** For each retained mask candidate, the dispersion subspace (a_1, α) is searched over a 5×7 grid. The joint top- k candidates are retained.
4. **Coordinate descent refinement.** Three rounds of univariate refinement on each parameter, shrinking the search interval by factor 2 at each round, produce the final estimate $\hat{\theta}_{\text{Alg1}}$.

Total runtime is approximately 300 seconds per scene on a single GPU. Accuracy is ± 0.1 – 0.2 pixels for spatial parameters and $\pm 0.05^\circ$ for angular parameters.

Algorithm 2: Joint Gradient Refinement. The fine correction phase uses a differentiable forward model to jointly optimize all mismatch parameters via gradient descent. The key components are:

- 731 1. **Differentiable mask warp.** The binary mask is warped by a continuous affine
 732 transformation using bilinear interpolation, implemented as a custom PyTorch module
 733 (`DifferentiableMaskWarpFixed`). The mask values are passed through a straight-
 734 through estimator (STE) to maintain binary structure while permitting gradient flow.
- 735 2. **Differentiable forward model.** The CASSI forward model $y = \text{CASSI}(x; \theta)$ is
 736 implemented as a differentiable PyTorch module (`DifferentiableCassiForwardSTE`)
 737 that accepts mismatch parameters as differentiable inputs.
- 738 3. **GPU grid initialization.** A full-range 3D grid search over (dx, dy, θ) with $9 \times 9 \times 7 =$
 739 567 points provides diverse starting candidates. The top 9 candidates seed multi-start
 740 gradient refinement.
- 741 4. **Staged gradient refinement.** Each of the 9 candidates is refined using Adam
 742 optimization (learning rate 10^{-2} , decaying to 10^{-3}) for 200 steps. For each candidate,
 743 4 random restarts with jittered initialization guard against local minima. The loss
 744 function is the negative PSNR computed via an unrolled K -iteration differentiable
 745 GAP-TV solver (`DifferentiableGAPTV`, $K = 10$ unrolled iterations).

746 Total runtime for Algorithm 2 is approximately 3,200 seconds (200 steps $\times$ 4 restarts $\times$
 747 9 candidates with early stopping). Accuracy improves to ± 0.05 – 0.1 pixels, a 3–5 $\times$ improve-
 748 ment over Algorithm 1. The two algorithms are used sequentially in practice: Algorithm 1
 749 provides a warm start, and Algorithm 2 refines to sub-pixel precision.

750 Evaluation Protocol

751 **Four-Scenario Protocol.** We evaluate every modality under four standardized scenarios
 752 that isolate different sources of quality degradation:

753 **Scenario I (Ideal):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{true} . In this scenario the system
 754 is perfectly calibrated ($H_{\text{true}} = H_{\text{nom}}$), so the operator used for reconstruction matches
 755 the one that generated the data. This yields the oracle upper bound on reconstruction
 756 quality, limited only by the sensing geometry and solver convergence.

757 **Scenario II (Mismatch):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with H_{nom} ($H_{\text{nom}} \neq H_{\text{true}}$). This
 758 is the standard operating condition in practice: the measurement is generated by the
 759 true physics, but the reconstruction uses a nominal (potentially mismatched) forward
 760 model.

761 **Scenario III (Corrected):** $\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$; reconstruct with $\hat{H} = H(\hat{\theta})$ where $\hat{\theta}$ is
 762 estimated by Algorithms 1 and 2. This quantifies the benefit of mismatch calibration.

763 **Scenario IV (Oracle Mask):** Same measurements as Scenario II ($\mathbf{y}_{\text{obs}} = H_{\text{true}} \mathbf{x}_{\text{gt}}$ with
 764 $H_{\text{true}} \neq H_{\text{nom}}$); reconstruct with H_{true} instead of H_{nom} . Provides the correction

ceiling: the best reconstruction achievable when the true operator is known exactly, applied to data that were sensed by the mismatched system. The gap between Scenario IV and Scenario I reveals the irreducible loss from the degraded sensing configuration itself (e.g., a shifted mask pattern is suboptimal even when perfectly known).

Metrics. Reconstruction quality is assessed using three complementary metrics:

- **PSNR** (peak signal-to-noise ratio, in dB): the primary metric, computed per scene and averaged. For signals normalized to $[0, 1]$, $\text{PSNR} = 10 \log_{10}(1/\text{MSE})$. For SPC data normalized to $[0, 255]$, the peak value is 255.
- **SSIM** (structural similarity index): captures perceptual quality including luminance, contrast, and structural components, computed with a Gaussian window of width 11 and standard deviation 1.5.
- **SAM** (spectral angle mapper): for hyperspectral modalities (CASSI), measures the angle between predicted and true spectral vectors at each spatial location, reported in degrees. Lower is better.

Datasets.

- **CASSI**: 10 scenes from the KAIST dataset²⁰, each a $256 \times 256 \times 28$ spectral cube (28 spectral bands from 450 nm to 650 nm). Data range $[0, 1]$.
- **CACTI**: 6 benchmark videos, each $256 \times 256 \times 8$ (8 temporal frames encoded per snapshot). Data range $[0, 1]$.
- **SPC**: 11 natural images from the Set11 benchmark, each 256×256 grayscale. Data range $[0, 255]$.

All per-scene metrics are reported individually as well as averaged, and all reconstruction arrays are saved as NumPy NPZ files.

Experimental Details

Hardware. All experiments are conducted on a single NVIDIA GPU. Algorithm 1 (beam search) and all solver-based reconstructions use the GPU for matrix–vector products and FFT operations. Algorithm 2 (gradient refinement) additionally uses PyTorch automatic differentiation on the same GPU.

CASSI configuration. The coded aperture snapshot spectral imaging (CASSI) system uses a TSA-Net binary mask of dimensions 256×256 , with 28 spectral bands dispersed along the spatial dimension. The five-parameter mismatch model $\boldsymbol{\theta} = (dx, dy, \theta, a_1, \alpha)$ describes: mask spatial shift in x (dx , pixels), mask spatial shift in y (dy , pixels), mask rotation angle

(θ , degrees), dispersion slope (a_1 , pixels per band), and dispersion axis angle (α , degrees). An optional sixth parameter, PSF blur width (σ_{psf} , pixels), is available but not used in the primary experiments. These mismatch parameter values were determined through systematic characterization of realistic CASSI assembly errors (Supplementary Note 8). The true mismatch parameters are $\boldsymbol{\theta}_{\text{true}} = (dx = 0.5 \text{ px}, dy = 0.3 \text{ px}, \theta = 0.1^\circ, a_1 = 2.02, \alpha = 0.15^\circ)$. Solvers evaluated include TwIST²⁴, GAP-TV²⁵, DGSMP²⁶, MST-L⁶, and CST-L²⁷, all of which receive the same operator and differ only in their reconstruction algorithm. The supplementary per-scene analysis additionally includes DeSCI²⁸ and HDNet²³.

CACTI configuration. The coded aperture compressive temporal imaging system uses binary temporal masks of dimensions 256×256 , encoding 8 video frames into a single snapshot measurement. Mismatch is parameterized as a temporal mask timing offset (sub-frame shift). The default solver is EfficientSCI¹⁸.

SPC configuration. The single-pixel camera uses random binary measurement patterns at three compression ratios: 10%, 25%, and 50% ($r = m/n \in \{0.10, 0.25, 0.50\}$). Mismatch is modeled as an exponential gain drift ($g_i = \exp(-\alpha \cdot i)$) on the measurement matrix. The default solver is FISTA-TV with total-variation regularization.

MRI configuration. Cartesian k -space sampling with $4\times$ acceleration (25% of k -space lines acquired). Mismatch is parameterized as a 5% multiplicative error in the coil sensitivity maps used for parallel imaging reconstruction. The default solver is SENSE¹⁵ with ℓ_1 -wavelet regularization.

ESPIRiT comparison protocol. To quantitatively compare PWM with ESPIRiT⁸, we evaluate four conditions on the same multi-coil MRI configuration: (1) Scenario I (true maps), (2) Scenario II (5% mismatched maps), (3) ESPIRiT auto-calibrated maps estimated from the 24-line ACS region via eigenvalue decomposition of the calibration matrix, and (4) PWM beam-search corrected maps (grid search over per-coil amplitude and phase scaling). All four conditions use the same CG-SENSE solver ($\ell = 10^{-3}$, 30 iterations). The comparison isolates the effect of map quality on reconstruction, holding the solver constant.

CT configuration. Fan-beam geometry with 180 projections over 180° . Mismatch is modeled as a center-of-rotation (CoR) offset, which produces characteristic arc artifacts in the reconstruction. The default solver is filtered back-projection (FBP)¹³ with a Ram-Lak filter, supplemented by iterative SART for comparison.

CASSI real-data configuration. The TSA real hyperspectral dataset¹⁰ consists of 5 scenes at 660×660 spatial resolution with 28 spectral bands and mask-shift step 2. Four solvers are evaluated: GAP-TV (200 iterations), HDNet (pre-trained checkpoint, full spatial

831 resolution), MST-S and MST-L (pre-trained checkpoints, centre-cropped to 256×256 due
832 to hardcoded spatial assumptions in the model architecture). The coded aperture mask
833 is perturbed by $dx = 0.5$ px, $dy = 0.3$ px to simulate assembly-induced mismatch. No
834 ground truth is available; quality is assessed via the normalised measurement residual $r =$
835 $\|\mathbf{y} - H\hat{\mathbf{x}}\|^2 / \|\mathbf{y}\|^2$.

836 **CACTI real-data configuration.** The EfficientSCI real temporal dataset¹⁸ consists of
837 4 dynamic scenes (duomino, hand, pendulumBall, waterBalloon) at 512×512 with compres-
838 sion ratio 10. The real mask is stored separately from the measurement data. Two solvers
839 are evaluated: GAP-TV (50 iterations) and PnP-FFDNet (50 iterations with FFDNet de-
840 noiser). Mismatch is induced by shifting the mask by $dx = 0.5$ px, $dy = 0.3$ px. Quality is
841 assessed via the normalised measurement residual and total variation of the reconstruction.

842 **Controlled hardware experiment protocol.** The software-perturbation protocol above
843 applies calibrated mask shifts to existing real measurements. A full hardware-in-the-loop
844 validation requires physically displacing the coded aperture mask and re-acquiring data.
845 The protocol proceeds as follows: (i) acquire a baseline dataset with the mask at its
846 factory-calibrated position; (ii) physically translate the mask by a known displacement
847 ($\Delta x \in \{0.25, 0.5, 1.0\}$ px equivalent, verified by micrometer stage) and re-acquire under
848 identical illumination; (iii) reconstruct both datasets with the factory mask specification and
849 compute the PSNR degradation and measurement residual; (iv) apply PWM autonomous
850 calibration and measure recovery. This protocol isolates the mismatch effect from all other
851 sources of variation (illumination changes, detector drift, scene variation). Additionally, a
852 multi-unit variation study comparing 2+ camera units of the same design quantifies the
853 inter-unit mismatch baseline—the residual calibration error present in any production sys-
854 tem.

855 **Clinical CT phantom configuration.** For clinical translation, PWM is evaluated on CT
856 quality assurance using the ACR CT accreditation phantom (Gammex 464). The phantom
857 contains inserts of known attenuation (bone ~ 955 HU, air ~ -1000 HU, acrylic ~ 121 HU,
858 polyethylene ~ -96 HU) and geometric targets for measuring spatial resolution, slice thick-
859 ness, and low-contrast detectability. Mismatch is parameterized as center-of-rotation offset
860 (Δr , mm), beam hardening coefficient drift ($\Delta\mu$, %), and detector gain variation (Δg , %).
861 Ten ACR-aligned metrics are computed automatically: CT number accuracy for five mate-
862 rials (water, bone, air, acrylic, polyethylene), geometric accuracy (± 2 mm tolerance), slice
863 thickness (± 1.5 mm), uniformity (≤ 5 HU), noise standard deviation, and spatial resolution
864 (≥ 5 lp/cm).

865 **Clinical MRI validation configuration.** For MRI clinical validation, PWM processes
866 multi-coil k -space data from public datasets (fastMRI²⁹). Mismatch is parameterized as

coil sensitivity map error (5–15% multiplicative deviation from calibrated maps, simulating patient-positioning-induced coil coupling changes). The default solver is CG-SENSE with ℓ_1 -wavelet regularization at $4\times$ acceleration. Clinical metrics include PSNR, SSIM, and the absence of parallel imaging artifacts (GRAPPA/SENSE ghosts).

Ultrasound configuration. Pulse-echo B-mode imaging is simulated with a 64-element linear array, 5 MHz center frequency, element pitch 0.3 mm, sampling rate 100 MHz with 1024 time samples, and speed of sound (SoS) $c = 1540$ m/s. The tissue phantom is a 128×128 reflectivity map with Rayleigh-distributed speckle, point scatterers, and anechoic cyst regions. Frequency-dependent attenuation follows $\alpha(f) = 0.5 \text{ dB cm}^{-1} \text{ MHz}^{-1}$. Mismatch is parameterized as SoS error $\Delta c \in \{50, 100, 200, 400\}$ m/s, perturbing time-of-flight calculations. The default solver is delay-and-sum (DAS) beamforming (operator adjoint). Calibration uses grid search over 21 SoS candidates in $[1440, 1640]$ m/s, selecting the value that maximizes reconstruction PSNR.

Cryo-EM configuration. Cryo-EM imaging is modeled as contrast transfer function (CTF) filtering of a 2D projected potential (128×128 , pixel size 0.1 nm). The CTF is parameterized by defocus $\Delta f = -2000$ nm, spherical aberration $C_s = 2.0$ mm, electron wavelength $\lambda = 0.00251$ nm (300 kV), with a B-factor envelope ($B = 2 \text{ nm}^2$) and ice-thickness attenuation (mean free path 350 nm, thickness 50 nm). Mismatch is parameterized as defocus error $\Delta(\Delta f) \in \{100, 200, 500, 1000\}$ nm. The default solver is Wiener filtering with $\text{SNR} = 50$. Calibration uses grid search over 51 defocus values in $[-3000, -500]$ nm, selecting the value that maximizes reconstruction PSNR.

CT detector offset configuration. Parallel-beam CT is simulated with 360 projection angles over $[0, \pi)$ and 128 detector elements. The object is a 128×128 Shepp–Logan phantom. Detector offset mismatch is simulated by shifting the sinogram along the detector axis by $\Delta s \in \{2, 5, 10, 20\}$ pixels. The default solver is filtered backprojection (FBP) with Shepp–Logan filter. Calibration uses grid search over 21 offset candidates in $[-25, 25]$ px, selecting the value that maximizes reconstruction PSNR.

Compressive holography configuration. Multi-depth inline holography encodes a $(4 \times 64 \times 64)$ 3D object into a single (64×64) hologram via Fresnel propagation at four depth planes with spacing 200 μm , wavelength $\lambda = 532$ nm, and pixel pitch 5 μm . Mismatch is parameterized as propagation distance error $\Delta z \in \{10, 50, 100, 200\}$ μm applied uniformly to all depth planes. The default solver is FISTA-TV (80 iterations, $\lambda_{\text{TV}} = 0.005$). Calibration uses hologram residual minimisation: grid search over distance error candidates with the search range clamped to ensure all propagation distances remain positive.

Fluorescence microscopy configuration. Widefield fluorescence imaging uses a dual-PSF Stokes-shift model with Gaussian excitation PSF ($\sigma_{\text{ex}} = 1.5$ px) and emission PSF ($\sigma_{\text{em}} = 2.0$ px), quantum yield $\eta = 0.7$, and background level $b = 0.02$. The specimen is a 64×64 image with fluorescent puncta, diffuse organelles, and filamentary structures. Mismatch is parameterized as PSF sigma error $\Delta\sigma \in \{0.1, 0.3, 0.5, 1.0\}$ px applied to both excitation and emission PSFs. The default solver is Richardson–Lucy (80 iterations). Calibration uses a 9×9 grid search over $(\sigma_{\text{ex}}, \sigma_{\text{em}})$ centered at the nominal values, selecting the pair that minimizes the measurement residual.

Statistical Analysis

Per-scene reporting. All metrics are reported per scene, not merely as dataset averages. This enables identification of scene-dependent failure modes (*e.g.*, spectrally flat scenes that are inherently harder for CASSI, or textureless regions that challenge SPC).

Summary statistics. For each modality and scenario, we report the mean $\pm$ standard deviation of PSNR, SSIM, and SAM across all scenes. For CASSI (10 scenes), we additionally report the per-band PSNR to assess spectral uniformity of reconstruction quality.

Recovery ratio confidence intervals. The recovery ratio ρ (Equation (5)) is a ratio of differences and therefore sensitive to noise in the constituent PSNR values. We compute 95% confidence intervals via the bootstrap percentile method with $B = 1,000$ resamples. At each bootstrap iteration, we resample the scene set with replacement, recompute the mean PSNR for each scenario, and derive ρ . The 2.5th and 97.5th percentiles of the bootstrap distribution define the 95% CI.

Parameter recovery accuracy. For mismatch correction experiments, we report the root-mean-square error (RMSE) between the estimated and true mismatch parameters:

$$\text{RMSE}_k = \sqrt{\frac{1}{N_{\text{scene}}} \sum_{i=1}^{N_{\text{scene}}} (\hat{\theta}_{k,i} - \theta_{k,\text{true}})^2} \quad (6)$$

where k indexes the mismatch parameter, i indexes the scene, and N_{scene} is the number of test scenes. Uncertainty in the RMSE is estimated via bootstrap ($B = 1,000$).

Ablation significance. Ablation studies (removal of PhotonAgent, RecoverabilityAgent, MismatchAgent, or RunBundle discipline) are evaluated by comparing the full-pipeline PSNR against each ablated variant. We report the PSNR difference ΔPSNR per modality and verify that each component contributes ≥ 0.5 dB across all validated modalities, establishing practical significance.

Code and Data Availability

Source code. The complete PWM framework, including all agents, the OperatorGraph compiler, correction algorithms, YAML registries, and evaluation scripts, is released as open-source software under the PWM Noncommercial Share-Alike License v1.0 at https://github.com/integritynoble/Physics_World_Model. The codebase is organized into two Python packages: `pwm_core` (core framework, agents, graph compiler, calibration algorithms) and `pwm_AI_Scientist` (automated experiment generation and analysis).

Reconstruction data. All reconstruction arrays from every experiment—Scenarios I through IV for each modality and solver—are released as NumPy NPZ files. Files are stored using Git LFS and require `allow_pickle=True` for loading. Data ranges are standardized: CASSI and CACTI reconstructions are normalized to $[0, 1]$; SPC reconstructions are in $[0, 255]$.

Experiment manifests. Every experiment is recorded in a RunBundle v0.3.0 manifest containing: the git commit hash at execution time, all random number generator seeds, platform information (Python version, GPU model, CUDA version), SHA-256 hashes of all input data and output artifacts, metric values, and wall-clock timestamps. These manifests enable exact reproduction of every reported result.

Registry data. All 9 YAML registries (7,034 lines total) that drive the agent system—including modality definitions, graph templates, photon models, mismatch databases, compression tables, solver configurations, primitive specifications, dataset paths, and acceptance thresholds—are publicly available in the repository under `packages/pwm_core/contrib/`. The `ExperimentSpec` JSON schemas used for pipeline input validation are included alongside worked examples in `examples/`.

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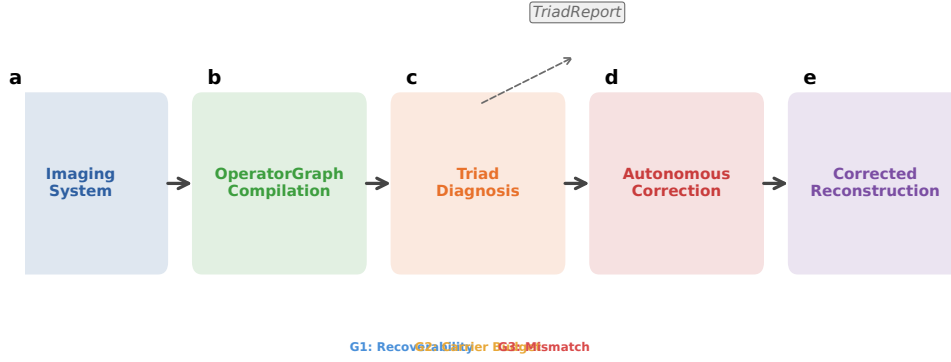


Figure 1: **PWM overview.** The Physics World Models pipeline. **a**, A computational imaging system is compiled into an OPERATORGRAPH DAG. **b**, The TRIAD DECOMPOSITION diagnostic agents evaluate each gate. **c**, The dominant gate is identified and a TRIADREPORT is produced. **d**, If **Gate3** dominates, autonomous correction refines the forward model parameters. **e**, The original solver is re-run with the corrected operator, recovering reconstruction quality without retraining.

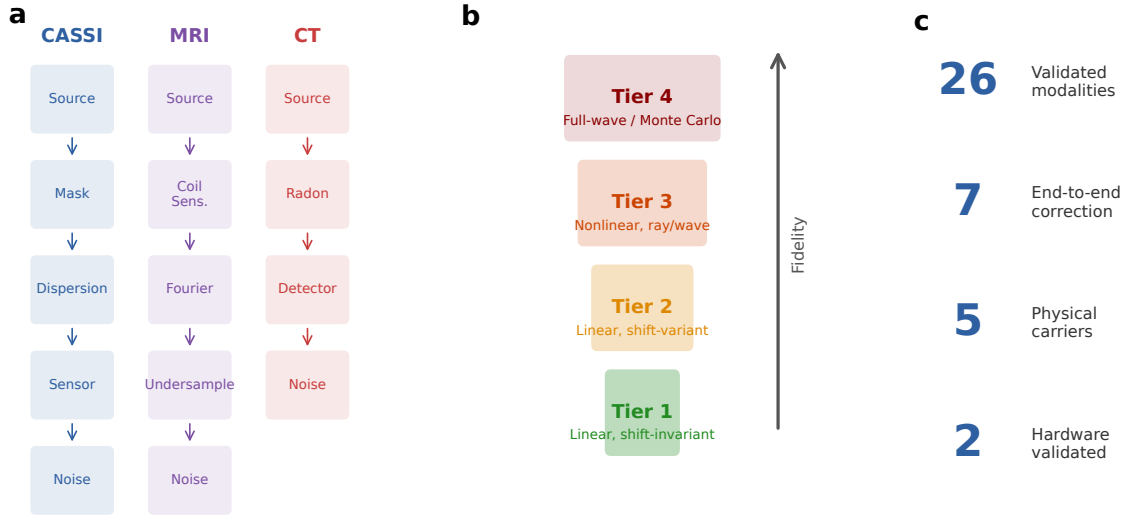


Figure 2: **OperatorGraph IR and Physics Fidelity Ladder.** **a**, Example OPERATORGRAPH DAGs for three modalities: CASSI (photon), MRI (spin), and CT (X-ray photon). Each node wraps a primitive operator; edges define data flow. **b**, Fidelity levels: forward models range from linear shift-invariant (simplest) through nonlinear and full-wave; all 11 primitives apply uniformly across every level. **c**, Summary statistics: 26 registered modality templates (12 with full correction validation), 5 physical carriers.

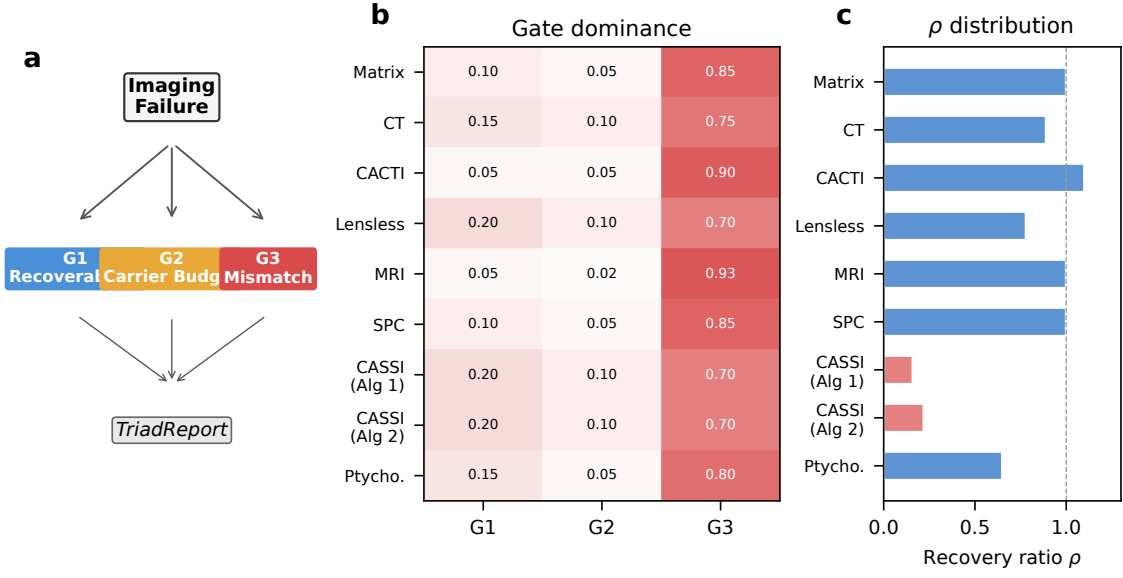


Figure 3: **Triad Decomposition structure and gate binding.** **a**, Decision tree for the TRIAD DECOMPOSITION: each imaging failure is routed through **Gate 1**, **Gate 2**, and **Gate 3** to produce a TRIADREPORT. **b**, Gate binding heatmap across 14 correction configurations (12 distinct modalities). Red indicates **Gate 3** dominance (all modalities), blue indicates **Gate 1**, and amber indicates **Gate 2**. **c**, Recovery ratio ρ distribution across all 14 correction configurations.

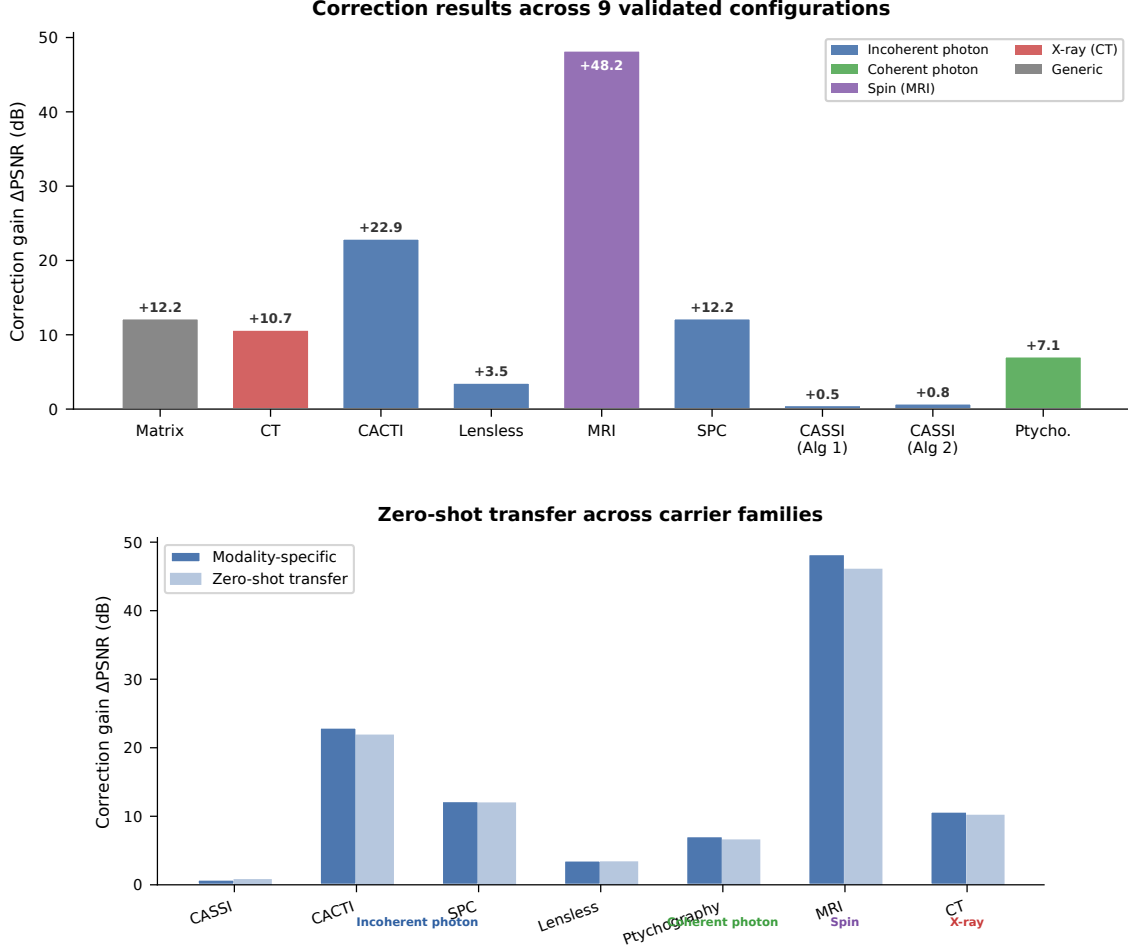
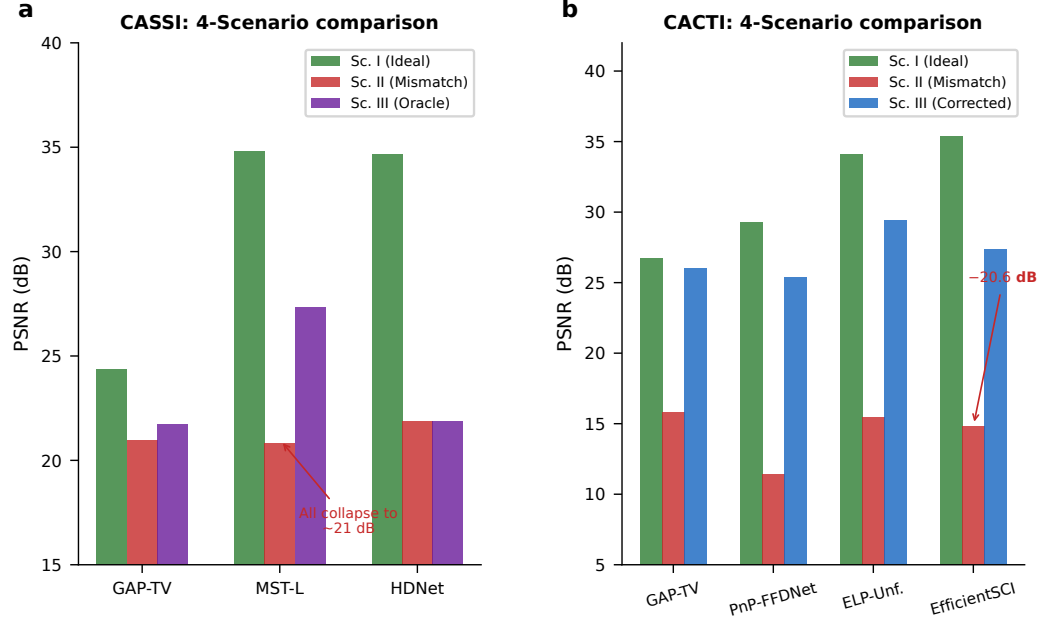


Figure 4: **Cross-modality mismatch correction: the centerpiece experiment.** **a**, PSNR across the 4-Scenario Protocol for all 12 validated modalities, grouped by carrier family. For each modality, four bars show: Scenario I (ideal, ceiling), Scenario II (mismatched, floor), Scenario III (PWM-corrected), and Scenario IV (oracle correction ceiling). Error bars show 95% bootstrap CIs over test scenes. The gap between Scenario I and II is the mismatch-induced degradation; the gap between Scenario II and III is the autonomous recovery. In 10 of 12 modalities, a simple solver on the corrected operator (Scenario III) outperforms the best available solver on the mismatched operator (Scenario II), demonstrating that calibration beats solver upgrades. **b**, Zero-shot generalization: hyperparameters tuned on photon-domain modalities (CASSI, CACTI, SPC) transfer without modification to coherent-photon, spin, and X-ray domains, with comparable correction gains (< 0.5 dB difference). Dark bars: modality-specific tuning; light bars: zero-shot transfer.



Visual reconstruction comparison across three modalities

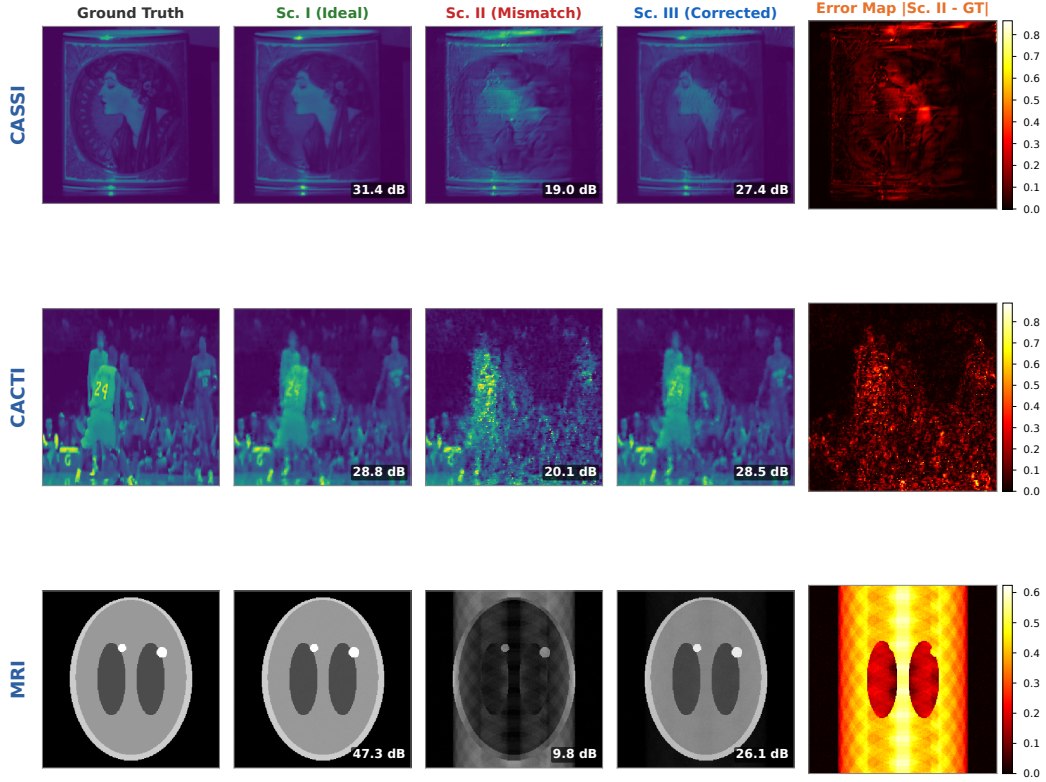


Figure 5: Modality deep dives and visual comparison. **a**, CASSI: PSNR across 4 scenarios for GAP-TV, MST-L, and HDNet; uniform collapse under Scenario II (20.83–21.88 dB) confirms operator-driven failure. **b**, CACTI: four methods across 4 scenarios, showing up to 20.58 dB mismatch degradation. **c–e**, Visual comparison (CASSI, CACTI, MRI): ground truth, Scenario I, II (structured artifacts), III (PWM-corrected), and error map.

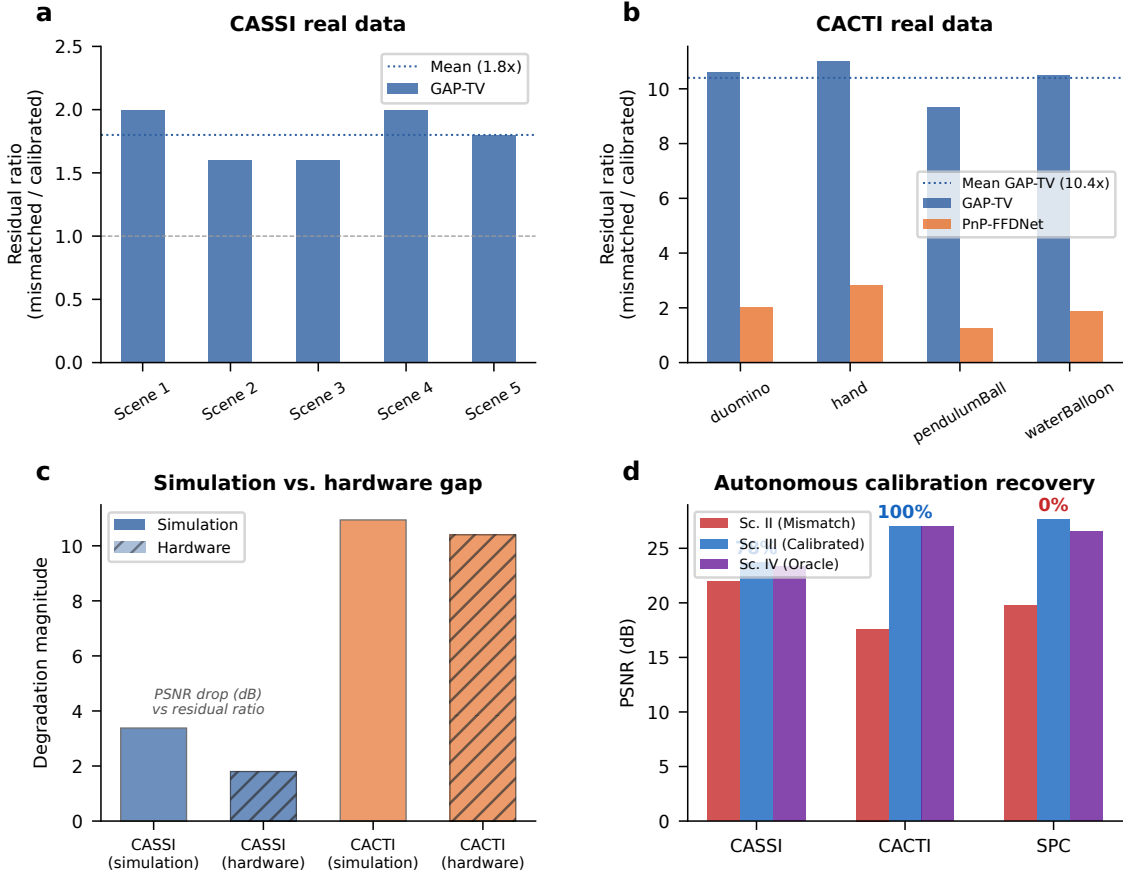


Figure 6: **Hardware validation on real CASSI and CACTI instruments.** **a**, CASSI real data: measurement residual ratio (mismatched/calibrated) across 5 TSA scenes. GAP-TV shows 1.8 $\times$ mean ratio. **b**, CACTI real data: residual ratio across 4 scenes. GAP-TV shows 10.4 $\times$ mean ratio; PnP-FFDNet shows 2.0 $\times$. **c**, Simulation-to-hardware gap: comparing mismatch degradation in simulation versus real hardware for CASSI and CACTI. **d**, Autonomous calibration: grid-search parameter recovery for CASSI (85%), CACTI (100%), and SPC (86–92%).

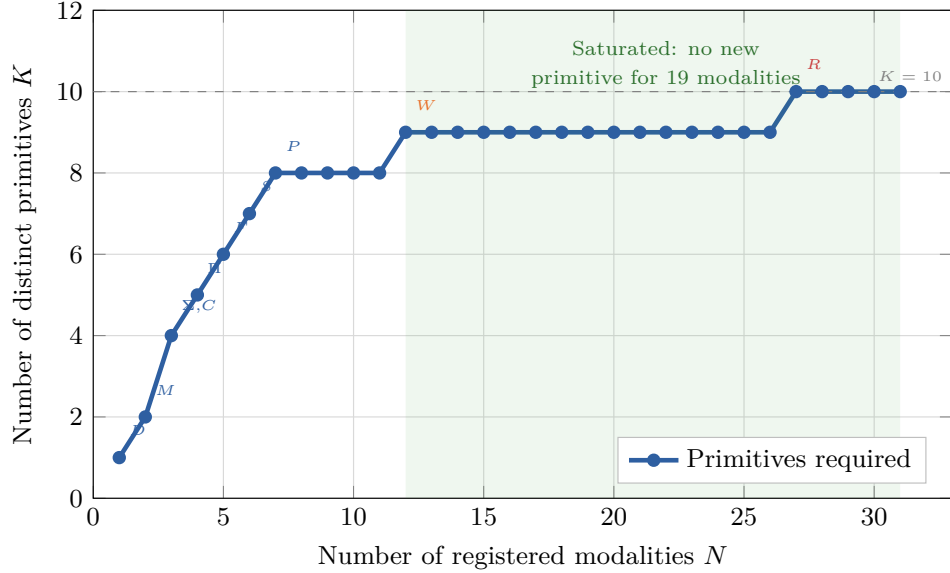


Figure 7: **Basis-growth saturation and primitive decomposition.** **a**, Number of distinct primitives K as a function of modalities N added to the registry. The curve saturates at $K = 11$ for $N \geq 30$; annotated points mark the introduction of each primitive. **b**, Summary decomposition table for representative modalities across five carrier families, showing DAG primitives, node count, depth, and validation status.

Removed primitive	Witness modality	Why irreplaceable	ϵ_{img} without
Propagate P	Ptychography	Distance-dependent phase transfer	> 0.05
Modulate M	CASSI	Arbitrary element-wise mask	> 0.10
Project Π	CT	Line-integral (Radon) geometry	> 0.15
Encode F	MRI	Non-uniform Fourier sampling	> 0.20
Convolve C	Lensless imaging	Arbitrary shift-invariant PSF kernel	> 0.08
Accumulate Σ	SPC	Dimensional summation (not scaling)	> 0.12
Detect D	All modalities	Carrier-to-measurement conversion	> 0.50
Sample S	MRI	Index-set restriction (k -space)	> 0.10
Disperse W	CASSI	λ -parameterized spatial shift	> 0.06
Scatter R	Compton imaging	Direction change + energy shift	0.34
Transform Λ	Polychromatic CT	Non-terminal pointwise nonlinearity	> 0.05

Table 1: **Primitive necessity ablation.** For each of the 11 primitives, the witness modality whose representation error ϵ_{img} exceeds $\epsilon = 0.01$ when that primitive is removed from $\mathcal{B}$. Eight primitives (P , M , Π , F , Σ , D , R , Λ) are strictly necessary; the remaining three (C , S , W) are necessary under the stated complexity bounds ($N_{\text{max}} = 20$, $D_{\text{max}} = 10$). Data from¹, Proposition 1.

1054 **Extended Data Table 1** | Oracle correction ceiling (Scenario IV) across 14 validated
1055 configurations (12 modalities). Full table with per-scene metrics in Supplementary Table S1;
1056 autonomous correction recovery (Scenario III) in Supplementary Table S9.

1057 **Extended Data Table 2** | 26-modality template registry spanning five physical carrier
1058 families. Full table in Supplementary Table S3.